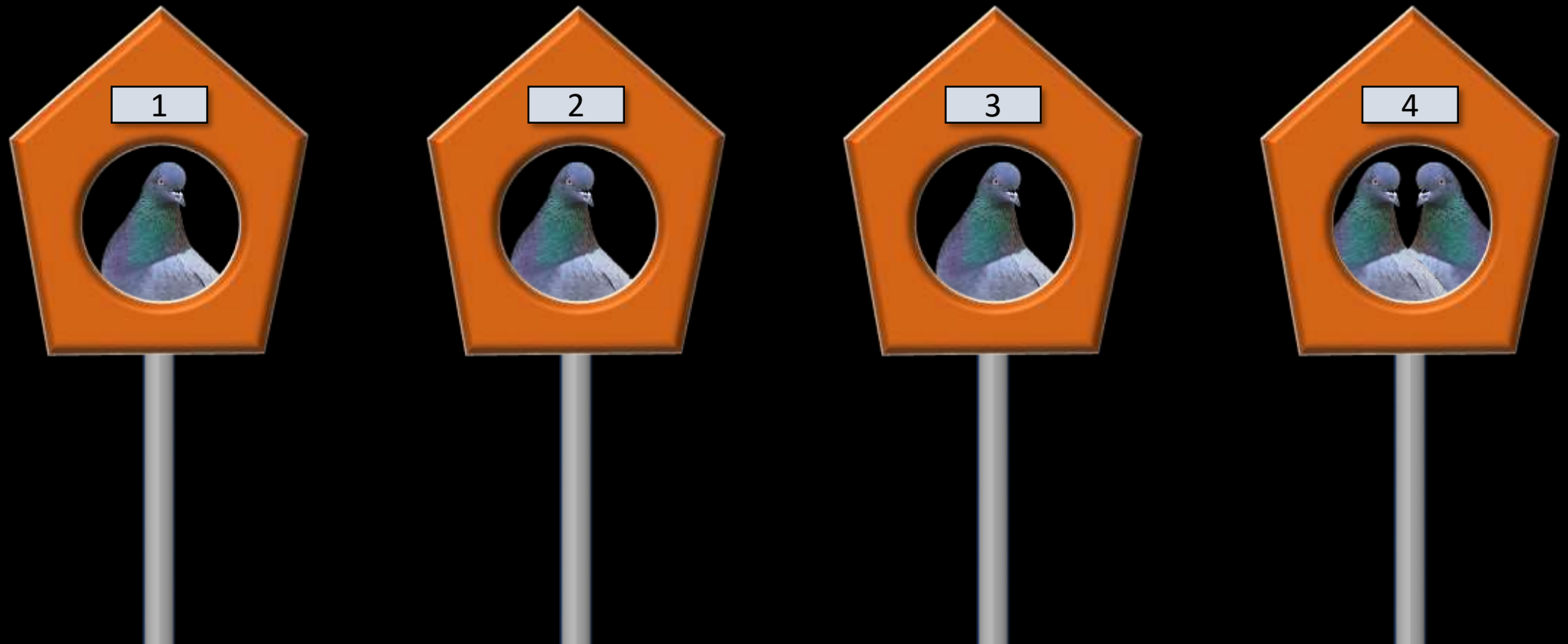
COUNTING PRINCIPLES AND PROBABILITY

Pigeonhole Principle

if n pigeons are put into m pigeonholes with $n>m$, then at least one pigeonhole must contain more than one pigeon



Counting Principle

1. Sum Rule – Suppose that an event can be performed by either of two different procedures, with *m* possible outcomes for the first procedure and *n* possibilities for the second. If the two sets of possible outcomes are disjoint, then, the number of possible outcomes for the event is *m+n*.
2. Product Rule – In a sequence of n_1 events in which the first has m possibilities and the second event has n_2, and the third event has n_3, and so forth, the total number of possibilities of the sequence will be $n_1(n_2)(n_3)\ldots(n_k)$.

Counting Principle

3. Factorial Notation – $n!$
4. Permutation – is an **arrangement** of all or part of a number of things in a **definite order**. The number of permutations of n objects taken r at a time is given by:
   - Permutation
     $$P = \frac{n!}{(n-r)!}$$
   - Permutation with Repeated Elements
     $$P_n = \frac{n!}{n_1!(n_2!)(n_3!)\ldots}$$ where $n_1+n_2+n_3 = n$
   - Circular Permutation
     $P_c = (n\text{-}1)!$

Counting Principle

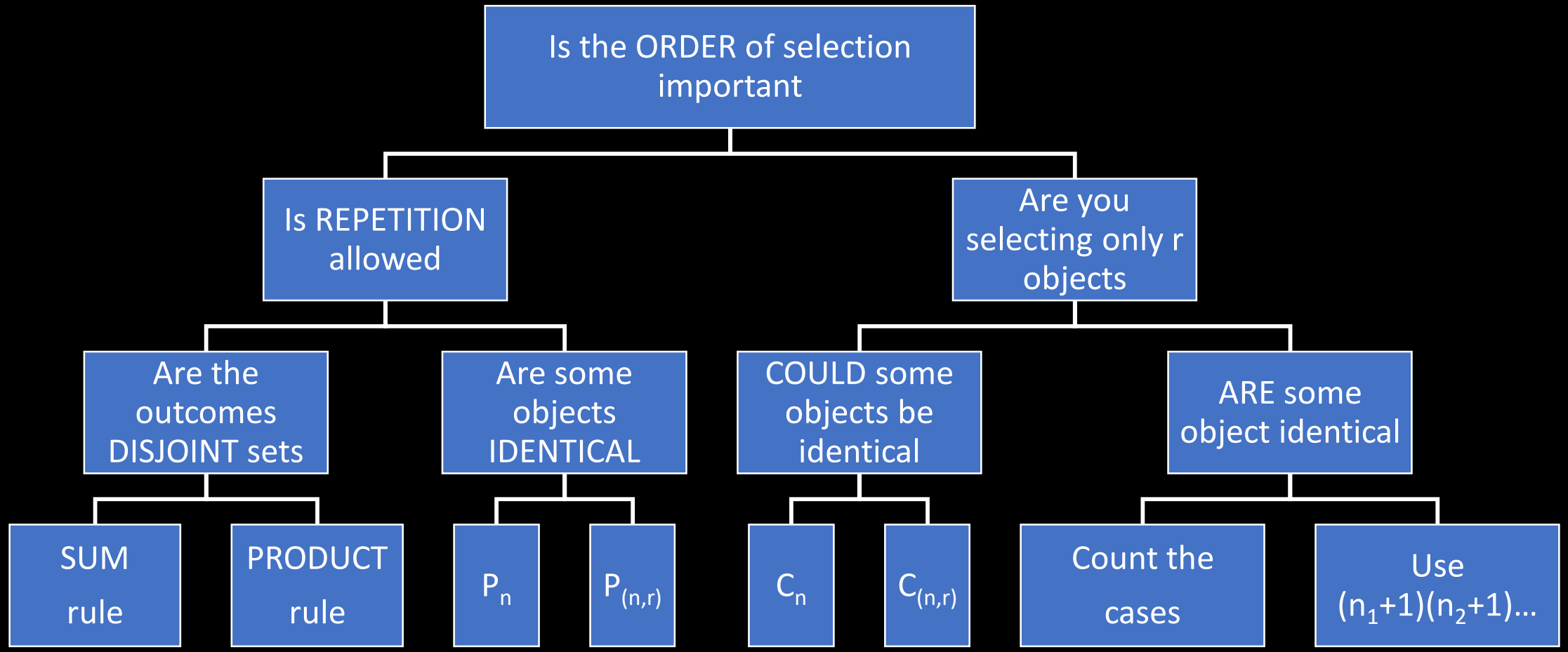
5. Combination – is a grouping or selection of all or part of a number of things (or objects) **without reference to the arrangement** of the things selected. The number of combinations of n objects taken r at a time is given by:

$$C(n,r) = \frac{n!}{(n-r)!\,r!}$$

- Combination of different things taken any number at a time

$$C_n = 2^n - 1$$

Counting Technique



Sample Space & Probability

A. Sample Space
   - Outcome
   - Sample space
   - Venn diagram / tree diagram

B. Simple and Compound Events
   - Event
   - Simple event (Elementary event)
   - Compound event (Composite event)

C. Probability
   - Classical
   - Empirical
   - Subjective